Document Title (placeholder — please replace with actual title)

CSS 132: Computer Programming for Engineers I

Overview

Instructor: Yusuf Pisan (pisan@uw.edu)

- Homepage || Research Group || All Course Evaluations || RateMyProfessors

- **Office hours:** M 1-2pm, Th 4-5pm @ UW1-260Q - signup via Canvas Calendar
- **Class Sessions:** TTh5:45-7:45 at UW2-141
- **Co-requisite:** CSSSKL 132
- Course website
- You can submit anonymous feedback for this class anytime during the quarter

This course is intended to be in-person, but we may need to switch to online instruction based on local community infection rates, campus infection rates, number of students in quarantine due to exposure, due to changes in UW or Washington State regulations or due to factors we cannot yet predict at this time. If any of the classes needs to be held online, you will be notified via your UW email address.

UW Mask mandate states all individuals are required to wear a face covering indoors, regardless of vaccination status, when on site at a University of Washington location. (20 Aug 2021)

What do I do if I have confirmed or suspected COVID-19? Contact the UW COVID-19 Response and Prevention Team. See COVID-19 frequently asked questions for details. The page also explains "What does the UW do when a member of our community has confirmed COVID-19?" In addition, you should install WA Notify — Washington Exposure Notifications app that lets you know when you've potentially been exposed to COVID-19.

Goals

Introduces programming concepts within social, mathematical, and technological context. Topics include
programming fundamentals (control structures, data types, functions, etc.), computer organization,
algorithmic thinking, introductory software engineering concepts, and social and professional issues.
Engineering applications are emphasized. A computer language used in engineering practice is used for
instruction. Co-requisite: CSSSKL 132.

Textbooks

zyBooks: Programming in C++, F Vahid and R Lysecky, https://www.zybooks.com/catalog/programming-c-plus-plus/

This is an e-book with integrated exercises. To get the e-book:

1. Sign in or create an account at https://learn.zybooks.com/ with your @uw.edu email

2. Enter zyBook code: UWBCSS132PisanFall2021

3. Subscribe

Class Sessions

We will be transitioning to a "flipped classroom" model for our instruction. To accomplish this you will need to complete readings and attempt exercises **before** the class session. Our in-person class sessions will focus on structured activities to practice the skills and ideas. This means the majority of your time seeing the course content will be on your own time outside the class session, while the class session will be to give you practice on the material you saw in the readings you completed earlier.

Motivation

In a "traditional classroom" model you go to a classroom and a teacher will lecture at you until the end of the session, and then you go home and do the hard work of actually trying to apply that material on practice problems or an assignment. Here are two things that are likely true about "traditional classroom" model:

- It is probably familiar to you.
- It is probably familiar to most of your instructors (it's how they learned, so it must work, right?).

Notice this list does not contain, "effective for student learning". Unfortunately for the "traditional classroom" model, most evidence has shown that it's quite ineffective of achieving that goal!

Think of learning programming or data science (or any skill for that matter) like learning how to ride a bike. How many people know how to ride a bike? Quite a few. How many of them do you think learned how to ride a bike by attending a lecture on bike riding?

Probably no one! Because that's obviously not learning how riding a bike works! To learn how to ride a bike, you have to actually go out there and ride the damn bike!

In more general terms, this means learning some concept or some skill **requires active participation** in the learning process from the students, most commonly accomplished by **deliberate practice** and **learning by doing**. Sitting passively in a lecture listening to someone talk fails at accomplishing this fundamental aspect of learning.

The flipped classroom model is built around fostering an environment for effective practice. Instead of coming to lecture to hear your instructor talk about some idea and then trying to go practice the idea at home (a much harder task), we flip this whole model around! You will read about the concept first (at your own pace) and then come to our class session to work with your classmates on practicing that skill! This has the added benefit of **collaborative learning** where you can work with your peers to construct a working knowledge of the things you saw from the lesson earlier.

Details

For every class session, there will be 3 components:

1. **Pre-CA**: Pre-Class Activities, readings, videos and exercises to complete **before** coming to class. If you do not complete these activities, you will not get the full benefit of the class time.

2. **ICA**: In-Class Activities. A combination of individual and group work to put into practice what you have learned. Some of the ICA will be graded, so it is important that you attend all classes and put in a good faith effort into the class activities. If you cannot attend the ICA, email me to let me know **before** class to be excused from the activities.

3. **Post-CA**: Post-Class Activities. Expanding on what we have covered in class and reflecting on what you have learned. You should complete these exercises soon after the class without waiting for the next class session.

Reflecting on what you have learned is necessary to solidify what you have learned. Think about the following questions

- Why did we learn this concept?
- How does this concept relate to everything we have seen so far?
- Can I describe this concept in my own words?
- Write down some "tips" for if you were going to help a colleague who was struggling with this content.

Research shows that using pen and paper to take notes is much more effective than using a computer. Computers are often distracting to you and also to others around you. I recommend keeping a dedicated notebook for this course and using the existing techniques for reflecting on content learned to help organize the material and your thoughts.

Expect to spend 10-15 hours per week outside class on average.

Inclusion

All students are welcome in CSS 132 and are entitled to be treated respectfully by both classmates and the course staff. I strive to create a challenging but inclusive environment that is conducive to learning for all students. If at any time you feel that you are not experiencing an inclusive environment, or you are made to feel uncomfortable, disrespected, or excluded, please report the incident so that we may address the issue and maintain a supportive and inclusive learning environment. You may contact me or the CSS Advisors to express your concerns. Should you feel uncomfortable bringing up an issue with a staff member directly, you may also consider sending anonymous feedback

Required Course Work

There will be three categories of required course work:

- **Exercises (15%)**

Short assignments and activities some to be performed in-class and some outside of class. Exercises will be graded as Complete/Incomplete. If you are making a good faith effort and submitting the exercise by the due date, you will receive a grade of Complete. Most students who continue to be engaged in the class have an exercise grade of 90-100%. The exercises help prepare you for the exams. Your 2 lowest exercise grades will automatically be dropped in Canvas.

If you are not able to attend class for any reason (personal, professional, Covid quarantine, mental health day, etc), you can email me **before the start of class time** to be exempt from that day's in-class exercises.

You can complete up to 5 bonus projects to increase your exercise grade. The report for the bonus project must be submitted within 1-week of completing the activity.

- **Homeworks (25%)**

We will have weekly homeworks. Most homeworks will be in the form of programming assignments, but will be more challenging and longer than exercises. Most students have a project grade of 80-100%. The most common reason for low project grade is not starting the project on time. Your lowest exercise grade will automatically be dropped in Canvas.

- **Exams (15% + 20% + 25%)**

We will have 2 exams plus a final exam. The exams are closed-book, closed-notes, except for a single cheat sheet (single sheet of letter-sized, double-sided notes created by you). In an exam, you have to demonstrate your knowledge in limited time without any additional resources. The average for exams is usually 70-75%.

Late assignment will only be accepted when it is due to exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% will receive a minimum of 2.0, 85% will receive a minimum of 3.0 and 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Getting Help

There are multiple ways of getting help:

- **Discord**

With an active class, discord can be one of the easiest and fastest ways to get help. Discord channel is an extension of our class and the same rules for in-class behavior apply to discord. I will also monitor the discord channel and contribute to the discussion from time to time. When asking questions:

- **Do not** post code.
- Be clear in stating the problem. Explain how you tried to solve it. Make sure the answer is not the first link that comes up in a Google search.
- You can post a warning/error message from compiler and ask how others have resolved it.
- You can ask how a specific C++ function gets used
- You can discuss your approach in high-level terms
- If you post a question and then figure out the answer, post your answer so it benefits others in the class

In general, if you can discuss the problem in English without having to show code, you can discuss it on Discord.

- **QSC**

Qualitative Skills Center has dedicated hours with 132 tutors that can help.

- **Office Hours**

Make an appointment to come and see me during office hours. I will not debug your code, but I will answer questions and try to help you figure out the problem. I will always ask what you have already done so far to solve the problem, so come prepared.

If you have a personal matter, please visit me during office hours to talk.

- **Email**

Email is not an effective way to get help. Unless your question is of personal nature, it should be posted on discord. If you believe the question really needs my attention, you can tag me on discord.

Extenuating Circumstances

I recognize that our students come from varied backgrounds and can have widely-varying circumstances. If you have any unforeseen circumstances that arise during the course, please do not hesitate to contact me to discuss your situation. **The sooner I am made aware, the more easily I can provide accommodations.** While some amount of “productive struggle” is healthy for learning, you should ask me for help if you have been stuck on an issue for a very long time.

Life happens! While our focus is providing an excellent educational environment, our course does not exist in a vacuum. My ultimate goal is to provide you with the ability to be successful, and I encourage you to work with me to make that happen. Please don't suffer in silence.

Access and Accommodations

Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DRS at 425.352.5307 or rosal@uw.edu.

DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s) and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

For Our Veterans

Welcome! We at UW Bothell understand that the transition into civilian life can be challenging for our veteran students and we have many resources for any who may want to reach out for guidance or assistance. This includes our Vet Corp Navigator through the WDVA and our Student Veterans Association (SVA). Please contact Veteran Services at 425.352.5307 or rosal@uw.edu. For those of you needing more URGENT support, please call The Suicide Prevention Hotline 1.800.273.8258 or connect with the UWB CARE Team https://www.uwb.edu/studentaffairs/care-team.

Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UWs policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy].
Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form].

Common Course Policies for the School of STEM

See https://www.uwb.edu/stem/about/stem-policies for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity. Student Support Services, Wondering How to Address Faculty? etc. (P.S. I prefer to be addressed as ‘Professor Pisan’)